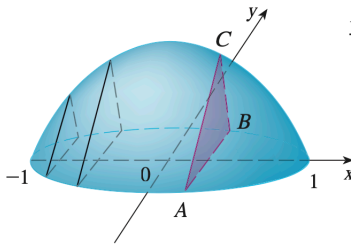
We now find the volumes of some solids which are *not* solids of revolution. In this case, we use the slicing method to calculate the volume as

$$V = \int_a^b A(x)dx,$$

where $A(x)$ is the cross-sectional area, which may not be a rotationally-symmetric.

1. The figure below shows a solid with a circular base of radius 1. Parallel cross-sections perpendicular to the base are equilateral triangles. Find the volume of the solid.



2. Find the volume of a pyramid whose base is a square with side length $L$ and whose height is $h$.

3. A wedge is cut out of a circular cylinder of radius 4 by two planes. One plane is perpendicular to the axis of the cylinder. The other intersects the first at an angle of $30^\circ$ along a diameter of the circle. Find the volume of the wedge.